

MILDRED LAKE, AB, Canada

WMO: 712550

Lat: 57.0403N

Lon: 111.5592W

Elev: 1017

StdP: 14.16

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.5	-24.7	-35.9	0.7	-29.1	-31.1	1.0	-24.4	20.9	19.2	18.4	18.4	3.1	100	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	84.5	63.1	81.0	61.5	77.8	60.3	65.9	77.9	64.2	76.0	62.6	73.4	8.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	85.8	69.3	60.0	80.4	67.6	58.3	75.4	66.5	31.2	78.1	29.9	76.1	28.7	73.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.6	14.5	DB	-34.0	90.9	5.0	2.9	-37.6	93.0	-40.6	94.7	-43.4	96.3	-47.0	98.4	(4)
(5)			WB	-34.1	69.2	4.9	2.6	-37.6	71.1	-40.5	72.6	-43.2	74.1	-46.8	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	1.8	8.3	20.6	37.9	51.4	61.0	65.4	62.2	52.3	38.7	19.8	5.9		
	DBStd	25.07	16.24	14.16	15.09	10.87	9.09	6.61	5.47	6.41	7.91	8.67	11.83	15.11		
	HDD50	6748	1494	1167	910	377	90	5	0	2	67	363	907	1366		
	HDD65	10895	1959	1587	1375	813	427	152	62	130	385	817	1357	1831		
	CDD50	1489	0	0	0	14	134	335	479	379	136	12	0	0		
	CDD65	159	0	0	0	0	7	31	75	42	4	0	0	0		
	CDH74	1657	0	0	0	4	114	344	686	433	74	2	0	0		
	CDH80	395	0	0	0	0	25	77	172	105	16	0	0	0		
	WSAvg	6.8	5.7	6.0	6.8	7.8	7.6	7.1	6.8	6.7	7.1	7.3	6.6	5.8		
	PrecAvg	16.5	0.8	0.7	0.7	0.7	1.4	2.5	2.9	2.4	1.7	1.1	0.7	0.8		
	PrecMax	21.0	1.4	2.5	1.5	2.6	3.8	5.1	5.4	4.6	4.0	6.0	1.3	1.7		
(15) Precipitation	PrecMin	9.8	0.3	0.0	0.1	0.2	0.4	0.9	1.0	0.6	0.4	0.2	0.3	0.3		
	PrecStd	2.9	0.3	0.5	0.4	0.6	0.9	0.9	1.2	1.1	1.1	1.3	0.3	0.4		
	0.4%	DB	40.9	42.9	54.9	71.6	83.7	86.3	88.8	87.8	81.7	68.2	47.3	40.0		
	MCWB	36.1	36.2	41.6	51.7	58.8	62.4	65.5	65.0	61.3	53.9	40.4	35.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	36.0	37.5	48.8	64.3	76.7	81.8	84.6	82.9	74.8	59.6	41.7	35.3		
	MCWB	32.8	33.1	39.0	47.0	55.4	60.4	64.2	63.5	59.0	48.1	36.2	32.2			
	5%	DB	30.7	33.2	44.3	59.6	72.3	77.8	81.1	78.8	69.8	54.9	37.9	30.6		
	MCWB	28.4	30.1	36.5	44.9	53.2	58.5	62.4	62.1	55.9	45.9	34.1	28.6			
	10%	DB	23.9	28.6	40.1	55.0	67.8	74.3	77.7	75.1	65.3	50.8	34.8	26.0		
	MCWB	22.4	26.3	34.3	42.5	51.1	57.4	61.5	60.4	54.3	43.4	32.1	24.4			
	0.4%	WB	36.8	36.9	43.0	52.0	60.9	66.2	69.3	68.6	63.1	54.6	40.8	36.1		
	MCDB	40.1	42.1	53.5	70.8	77.2	79.0	82.4	81.6	78.1	65.4	46.3	39.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.9	33.5	39.4	48.1	57.4	63.0	66.3	65.6	60.2	49.6	37.0	32.3		
	MCDB	35.9	37.3	47.5	61.7	73.0	74.5	78.3	77.4	72.3	57.0	40.9	34.9			
	5%	WB	28.4	30.2	37.0	45.5	54.9	61.1	64.7	63.7	57.6	46.4	34.5	28.5		
	MCDB	30.5	32.9	43.7	57.9	68.4	71.7	76.1	75.1	66.8	53.6	37.4	30.6			
	10%	WB	22.5	26.4	34.6	43.1	52.5	59.4	63.2	61.9	55.4	43.9	32.2	24.4		
	MCDB	23.8	28.5	39.8	54.0	65.5	70.5	73.9	72.3	63.0	50.3	34.8	25.8			
	5% DB	MDBR	14.9	18.1	20.4	20.5	21.1	19.4	19.5	19.4	18.2	13.5	11.5	13.7		
	MCDBR	20.3	22.6	22.6	27.1	27.3	25.4	25.2	25.8	26.8	21.3	14.6	17.1			
(36) Mean Daily Temperature Range	MCWBR	17.4	18.6	14.7	14.1	11.8	9.7	9.5	10.2	12.8	11.9	10.9	15.0			
	5% WB	MCDDBR	20.4	22.1	21.6	26.0	24.4	21.2	21.7	22.9	23.8	20.0	13.6	17.3		
	MCWBR	17.6	18.3	14.4	14.2	11.3	9.3	9.5	10.2	12.4	11.8	10.7	15.4			
(40) Clear-Sky Solar Irradiance	taub	0.246	0.270	0.297	0.338	0.358	0.382	0.397	0.367	0.332	0.301	0.261	0.234			
	taud	2.229	2.278	2.291	2.364	2.383	2.328	2.345	2.400	2.497	2.534	2.362	2.161			
	Ebn at Noon	207	252	274	279	279	272	265	266	261	242	203	174			
	Edh at Noon	16	23	30	33	35	37	36	32	25	18	14	12			
(44) All-Sky Solar Radiation	RadAvg	210	521	1033	1503	1771	1795	1711	1406	961	496	208	132			
	RadStd	18	29	64	103	173	147	97	95	93	69	25	10			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+3.33	N/A	N/A	-1.04	-1.55	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	+2.93	N/A	-0.67	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page